APEKSHA MORAPPANAVAR

Software Engineer | 1 Year of Experience | C++ · Linux · Backend & Full-Stack Development

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Software Engineer with 1 year of experience in C++, Linux, and backend/full-stack development (React, Node.js, Express, MongoDB, REST APIs, JWT-based authentication), currently transitioning from embedded/automotive systems into backend and distributed systems engineering. Combines production C++ development with hands-on experience building and securing a full-stack, multi-tenant web application.

WORK EXPERIENCE

Software Engineer |*Magna Automotive India Pvt Ltd, Bangalore* March 2025 - Present

- Developed object-oriented C++ software components from system requirements, migrating legacy implementations to a modern architecture and improving maintainability and code readability.
- Designed and executed 50+ unit tests using Google Test (gTest), strengthening software correctness and supporting regression testing practices.
- Built 3+ Python automation tools for data processing, result visualization, and log conversion, reducing manual effort in validation workflows.
- Performed performance profiling and optimization, achieving ~40% improvement in runtime efficiency over baseline implementations.
- Collaborated with cross-functional engineering teams to deliver production-ready software, maintaining design, testing, and documentation standards.
- Performed static code analysis using QA-C++ to identify and resolve coding-standard violations and maintainability issues.

PROJECTS

MultiFlix — Full-Stack Multi-Tenant Media Platform |*Personal Project | GitHub*

- Built a full-stack media platform with a multi-tenant architecture where each billing account manages multiple users, roles, and isolated profiles.
- Implemented authentication and authorization using JWT and Passport.js, with role-based access control (RBAC) distinguishing owner and member permissions across tenant accounts.
- Developed profile-scoped personalization features — watch history, watchlist, continue-watching, and recommendation generation — sourced from TMDb data.
- Designed and integrated REST APIs for content discovery, Google OAuth login, and invite-based onboarding, with MongoDB/Mongoose-backed persistence on the backend.
- Built the frontend with React, TypeScript, Vite, and Tailwind CSS, and the backend with Node.js and Express.

Breast Cancer Detection Using Few-Shot Learning |*Research Project*

- Built a meta-learning-based model using Prototypical Networks and the BUSI dataset to classify ultrasound images into three classes: benign, malignant, and normal.
- Applied a CNN-based encoder with 4 layers, utilizing advanced data augmentation techniques such as image rotations of 90, 180, and 270 degrees, and multithreading to improve model performance with limited labeled data.
- Achieved classification accuracies of 85.75% to 91.67% across different k-shot (5, 10, and 15) configurations, outperforming traditional models like VGG-16 and Inception V3.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science and Engineering |*KLE Technological University* 2021 - 2025

CGPA: 9.44

TECHNICAL SKILLS

Backend & Systems: C / C++, Object-Oriented Programming (OOP), Data Structures & Algorithms, REST API Design, Multithreading, Distributed Systems, System Design, Linux, Git, Debugging, Optimization & Complexity Analysis, Unit & Integration Testing (Google Test)

Web / Full-Stack: JavaScript, TypeScript, HTML, CSS, React.js, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, Mongoose, JWT & Passport.js Authentication, Role-Based Access Control (RBAC), OAuth (Google), Web Development, Software Design & Architecture

ML / Data: Python, CNNs, Few-Shot Learning, Data Augmentation, Model Evaluation

Other Tools: MATLAB, ROS2, Helix QAC (static analysis), Oracle SQL, Informatica Power Center, Windows

ACHIEVEMENTS

- Participated in the Smart India Hackathon and won the internal hackathon round at the institute level.
- Presented research on prototypical networks and few-shot learning for disease detection at INCET 2024.

CERTIFICATIONS

- Linear Programming and Its Applications to Computer Science — IIT Kanpur
- Breast Cancer Detection by Prototypical Networks Using Few-Shot Learning — IEEE Bangalore
- JNCIA-Junos (Juniper Networks Certified Associate – Junos) — Juniper Networks